

$$\underline{13.1.3)} \quad \operatorname{res}_{z=1} \frac{e^z}{(z-1)^2} = \lim_{z \rightarrow 1} (e^z)' = e, \text{ v.k. } z=1 - \Pi^2$$

$$\underline{13.3.1)} \quad f(z) = \frac{e^{z+1}}{(z+1)^2}, \quad \operatorname{res}_{z=2} f(z) = -\operatorname{res}_{z=-1} f(z) = -\lim_{z \rightarrow -1} e^{z+1} = -1$$

$$\underline{13.5.5)} \quad f(z) = \sin z \sin \frac{1}{z} \quad \begin{cases} z=0 - \text{cor} \\ z=\infty - \text{cor} \end{cases}$$

$$\sin z \sin \frac{1}{z} = \left(z - \frac{z^3}{6} + \dots \right) \left(\frac{1}{z} - \frac{1}{6z^3} + \dots \right) \Rightarrow C_{-1} = 0 \Rightarrow$$

$$\Rightarrow \operatorname{res}_{z=\infty} f(z) = 0 \Rightarrow \operatorname{res}_{z=0} f(z) = 0$$

$$\underline{13.5.6)} \quad f(z) = \frac{\cos z}{(z^2+1)^2} \quad \begin{cases} z=\infty - \text{cor} \\ z=\pm i - \Pi^2 \end{cases}$$

$$\operatorname{res}_{z=i} f(z) = \lim_{z \rightarrow i} (f(z)(z-i)^2)' = \lim_{z \rightarrow i} \frac{-\sin z (z+i)^2 - \cos z 2(z+i)}{(z+i)^4} = \frac{-i}{4e}$$

$$\operatorname{res}_{z=-i} f(z) = \lim_{z \rightarrow -i} (f(z)(z+i)^2)' = \frac{i}{4e}$$

$$\operatorname{res}_{z=\infty} f(z) = \frac{i}{4e} - \frac{i}{4e} = 0$$

$$\underline{14.1.5)} \quad \oint_{|z|=3} \frac{z^4}{z^2+z-2} dz = 2\pi i \left(\frac{1}{3} - \frac{16}{3} \right) = -10\pi i$$

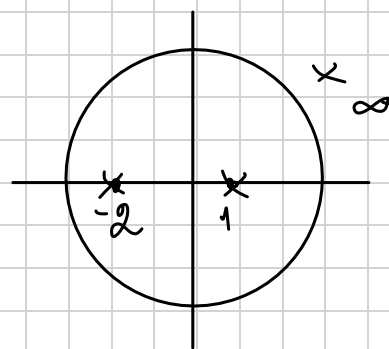
$$z=\infty - \Pi^4$$

$$z=-2 - \Pi^1$$

$$z=1 - \Pi^1$$

$$\operatorname{res}_{z=2} f(z) = \lim_{z \rightarrow 2} \frac{z^4}{z-1} = \frac{-16}{3}$$

$$\operatorname{res}_{z=1} f(z) = \lim_{z \rightarrow 1} \frac{z^4}{z+2} = \frac{1}{3}$$



$$\underline{14.2.3)} \quad \oint_{|z-\frac{1}{2}|=1} \frac{z^3 e^{1/z}}{1-z^2} dz = -2\pi i \left(1 + \frac{1}{2e} \right)$$

$$|z-\frac{1}{2}|=1$$

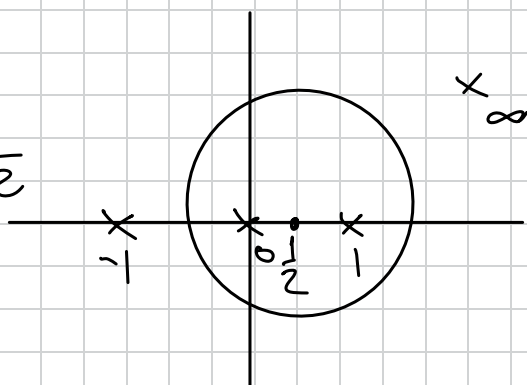
$$z=\infty - \text{cor}$$

$$z=\pm 1 - \Pi^1$$

$$z=0 - \text{cor}$$

$$\operatorname{res}_{z=-1} f(z) = \lim_{z \rightarrow -1} \frac{z^3 e^{1/z}}{1-z} = \frac{1}{2e}$$

$$f(z) = \frac{z^3 e^{1/z}}{1-z^2} =$$



$$= -z \frac{e^{1/z}}{1 - \frac{1}{z^2}} = -z \left(\sum_{k=1}^{\infty} \frac{1}{k!} \frac{1}{z^k} \right) \left(\sum_{n=1}^{\infty} \frac{1}{z^{2n}} \right) \Rightarrow \operatorname{res}_{z=\infty} f(z) = 1$$

$$8) \oint_{|z|=5/2} \frac{z^2}{z-3} \sin \frac{z}{z-2} dz = -2\pi i (98 \sin 3 - 10 \cos 1 - 7 \sin 1)$$

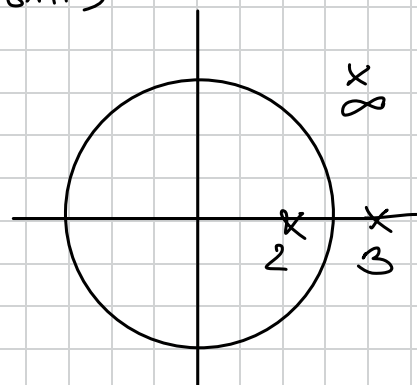
$|z|=5/2$

$$z = \infty - \cot$$

$$\operatorname{res}_{z=3} f(z) = 98 \sin 3$$

$$z = 3 - \eta^1$$

$$z = 2 - \cot$$



$$f(z) = z \frac{1}{1 - 3/z} \sin \frac{1}{1 - 2/z} = z \left(1 + \frac{3}{z} + \frac{9}{z^2} + \dots \right) \sin \left(1 + \frac{2}{z} + \frac{4}{z^2} + \dots \right) =$$

$$= z \left(1 + \frac{3}{z} + \frac{9}{z^2} + \dots \right) \left(\sin 1 \cos \left(\frac{2}{z} + \frac{4}{z^2} \right) + \cos 1 \sin \left(\frac{2}{z} + \frac{4}{z^2} \right) \right)$$

$$\text{mu } \frac{1}{z} : -28 \sin 1 + 4 \cos 1 + 6 \cos 1 + 9 \sin 1 = 10 \cos 1 + 7 \sin 1$$

$$17) \oint_{|z|=1} \frac{(z-i) \sin \frac{1}{z}}{(z-3i)^2} dz = -2\pi i \left(\frac{2}{9} \cos \frac{1}{3} - 8 \sin \frac{1}{3} \right)$$

$$z = 3i - \eta^2$$

$$\operatorname{res}_{z=3i} f(z) = \operatorname{res}_{z=3i} \left((z-i) \sin \frac{1}{z} \right)' = -\sin \frac{1}{3} + \frac{2}{9} \cos \frac{1}{3}$$

$$z = 0 - \cot$$

$$\operatorname{res}_{z=\infty} f(z) = \lim_{z \rightarrow \infty} z(f(\infty) - f(z)) = 0$$

$$z = \infty - \cot$$

$$22) \oint_{|z-\pi/2|=3} \frac{z - i \ln 2}{z^2 - i/3} dz = 2\pi i \cdot \frac{9}{4} \sqrt{2} = \frac{9}{2} \pi^2 i$$

$|z - \frac{\pi}{2}| = 3$

$$z = \pi k + i \frac{1}{2} \ln 2 - \eta^1$$

$$z = \pi/2 + \pi k - \cot$$

$$z = \pi/2 - \eta^1$$

$$z = \frac{i}{2} \ln 2$$

$$z = \pi + i \frac{1}{2} \ln 2 - \eta^1$$

Geben keine Rückmeldung:

$$\operatorname{res}_{z=\pi/2} f(z) = (z - \pi/2) \frac{z - i \ln 2}{z^2 - i/3} \Big|_{z=\pi/2} = 0$$

$$\operatorname{res} f(z) = \left(z - \frac{i}{2} \right) \frac{z - i \ln 2}{\log z - 1/3} \Big|_{z = \frac{i}{2} \ln 2} = 0 \quad (\text{yes})$$

$$\operatorname{res} f(z) = \frac{z - i \ln 2}{-1/\cos^2 z} = \frac{2\pi}{8/9} = \frac{9}{4}\pi$$

$$\underline{14.3.2)} \int_0^{2\pi} \frac{d\varphi}{(a + b \cos \varphi)^2} = \oint_{|z|=1} \frac{-i dz}{z \left(a + \frac{b}{2} \left(z + \frac{1}{z} \right) \right)^2} = \oint \frac{-i z dz}{(az + \frac{b}{2}(z^2 + 1))^2} \quad \textcircled{=}$$

$$\text{Zunehm } z = e^{i\varphi} \quad \varphi \in (0, 2\pi] \quad dz = i e^{i\varphi} d\varphi = iz d\varphi$$

$$\cos \varphi = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$az + \frac{b}{2}(z^2 + 1) = 0 \Rightarrow z = \frac{-a \pm \sqrt{a^2 - b^2}}{b} \quad - \pi^2 \quad - \text{Faktor } c + \text{ nearest } b \text{ distance}$$

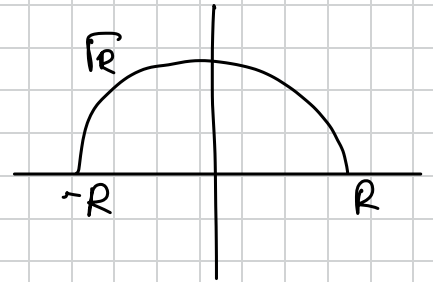
$$\operatorname{res} f(z) = \left(\frac{-4iz}{b^2(z - z_2)^2} \right)' \Big|_{z=z_1} = \frac{-ia}{(a^2 - b^2)^{3/2}} \Rightarrow \int \dots = \frac{2\pi a}{(a^2 - b^2)^{3/2}}$$

$$\underline{23.1.3)} \int_{-\infty}^{+\infty} \frac{dx}{x^2 - 2ix - 2} \quad \textcircled{=}$$

$$\int_{\Gamma} f(z) dz \rightarrow 0$$

$$z = i \pm 1 - \pi^1$$

$$\boxed{\operatorname{Im} z \geq 0}$$



$$\operatorname{res} f(z) = \frac{1}{z - (i-1)} \Big|_{i+1} = 1/2$$

$$\operatorname{res} f(z) = -1/2 \quad z = i-1$$

$$\textcircled{=} 2\pi i \left(\frac{1}{2} - \frac{1}{2} \right) = 0$$

$$6) \int_{-\infty}^{\infty} \frac{x^2 dx}{x^4 + 6x^2 + 25} \quad \textcircled{=} \quad f(z) = \frac{z^2}{z^4 + 6z^2 + 25} \Rightarrow \int_{\Gamma} f(z) dz \rightarrow 0$$

$$z = \pm 1 \pm 2i - \pi^2$$

$$\operatorname{res} f(z) = \frac{(1+2i)^2}{4i \cdot 2(2+4i)} = \frac{1+2i}{16i}$$

$$\operatorname{res} f(z) = \frac{(-1+2i)^2}{4i \cdot 2(2-4i)} = \frac{1-2i}{16i}$$

$$\textcircled{=} 2\pi i \left(\frac{1+2i}{16i} + \frac{1-2i}{16i} \right) = \pi/4$$

$$\underline{23.2.7)} \int_{-\infty}^{\infty} \frac{(2-x) \cos(3x-2)}{x^2-2x+2} dx = I$$

$$\mathcal{J} = \int_{-\infty}^{\infty} \frac{(2-z)e^{i(3x-2)}}{z^2-2z+2} dz \Rightarrow I = \operatorname{Re} \mathcal{J}$$

$$z = 1 \pm i - n', \text{ depend } 1 \pm i$$

$$\operatorname{res}_{1+i} f(z) = \frac{(1-i)e^{-2i}}{2i} e^{3i-3} = \frac{-1-i}{2e^3} (\cos$$

$$\text{No more residues for } f(z) = g(z)e^{i3z}, g(z) = \frac{(2-z)e^{-2i}}{z^2-2z+2}$$

$$\lim_{R \rightarrow \infty} \max_{z \in \Gamma_R} |g(z)| \rightarrow 0 \quad \Gamma_R = \{z: |z|=R, \operatorname{Im} z \geq 0\}$$

$$\mathcal{J} = 2\pi i \cdot \frac{-1-i}{2e^3} (\cos 1 + i \sin 1) \Rightarrow I = \frac{\pi}{e^3} (\cos 1 + \sin 1)$$

$$15) \int_{-\infty}^{\infty} \frac{x \sin(2-x)}{x^2+2} dx = I \quad \mathcal{J} = \int_{-\infty}^{\infty} \frac{-x e^{i(x-2)}}{x^2+2} dx \quad I = \operatorname{Im} \mathcal{J}$$

$$z = \pm i\sqrt{2} - n' \quad \operatorname{res}_{i\sqrt{2}} f(z) = \frac{-i\sqrt{2}e^{-2i}}{2i\sqrt{2}} e^{-i\sqrt{2}} = \frac{-1}{2e^{2i+i\sqrt{2}}}$$

$$\text{No more residues for } f(z) = g(z)e^{iz}, g(z) = \frac{-ze^{-2i}}{z^2+2}$$

$$\lim_{R \rightarrow \infty} \max_{z \in \Gamma_R} |g(z)| = 0 \Rightarrow \mathcal{J} = 2\pi i \left(\frac{-1}{2e^{2i+i\sqrt{2}}} \right) = \frac{-\pi}{e^{\sqrt{2}}} i (\cos 2 + i \sin 2) =$$

$$\Rightarrow I = \frac{-\pi}{e^{\sqrt{2}}} \cos 2$$

$$19) \int_{-\infty}^{\infty} \frac{x^3 \cos(1-2x)}{(x^2+1)(x^2+4)} dx = I \quad \mathcal{J} = \int_{-\infty}^{\infty} \frac{x^3 dx}{(2x^2+1)(x^2+4)} e^{i(2x-1)} \quad I = \operatorname{Re} \mathcal{J}$$

$$\left[\begin{array}{l} z = \pm i/\sqrt{2} \\ z = \pm 2i \end{array} \right. - n', \quad \operatorname{res}_{i/\sqrt{2}} f(z) = \frac{1}{4} e^{-i\sqrt{2}} (\cos 1 - i \sin 1)$$

$$\operatorname{res}_{2i} f(z) = \frac{2}{7} e^{-4} (\cos(-i \sin 1))$$

$$\text{No more residues for } f(z) = g(z)e^{2iz}, g(z) = \frac{x^3 e^{-i}}{(2x^2+1)(x^2+4)} \quad \lim_{R \rightarrow \infty} \max_{z \in \Gamma_R} |g(z)| = 0$$

$$I = 2\pi \left(\frac{2}{7} e^{-4} - \frac{1}{4} e^{-i\sqrt{2}} \right) \sin 1 = \frac{2\pi \sin 1}{7} \left(\frac{2}{e^4} - \frac{1}{2e^{\sqrt{2}}} \right)$$

$$\text{VI a) } \int_{-\infty}^{+\infty} \frac{\sin(2x+i)}{x^2-2x+5} dx = \int_{-\infty}^{+\infty} \frac{\sin 2x \cos i + \cos 2x \sin i}{x^2-2x+5} dx =$$

$$= \text{ch} \int \underbrace{\frac{\sin 2x}{x^2-2x+5}}_{I_1} dx + i \text{sh} \int \underbrace{\frac{\cos 2x}{x^2-2x+5}}_{I_2} dx$$

$$f = \int_{-\infty}^{+\infty} \frac{e^{2ix}}{x^2-2x+5} dx \quad I_1 = \text{Im} f \quad I_2 = \text{Re} f$$

$$\text{res}_{1+2i} f(z) = \frac{1}{4i} e^{2i-4} = \frac{i}{4e^4} (\cos 2 + i \sin 2)$$

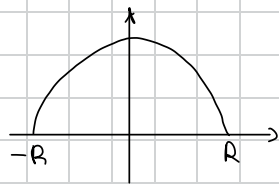
$$\text{No n. d. } f(z) = g(z) e^{2ix}, \quad g(z) = \frac{1}{z^2-2z+5}$$

$$\lim_{R \rightarrow \infty} \max_{z \in \Gamma_R} |g(z)| = 0 \Rightarrow I_1 = \frac{\pi}{2e^4} \sin 2 \quad I_2 = \frac{\pi}{2e^4} \cos 2$$

$$\int_{-\infty}^{+\infty} \frac{\sin(2x+i)}{x^2-2x+5} dx = \frac{\pi}{2e^4} (\text{ch} \sin 2 + i \text{sh} \cos 2)$$

$$\text{VI b) } \int_{-\infty}^{+\infty} \frac{\cos x}{x^2-4x+1-4i} dx = \int_{-\infty}^{+\infty} \frac{e^{ix}}{2(x^2-4x+1-4i)} dx + \int_{-\infty}^{+\infty} \frac{e^{-ix}}{2(x^2-4x+1-4i)} dx =$$

$$= \underbrace{\int_{-\infty}^{+\infty} \frac{e^{ix}}{2(x^2-4x+1-4i)} dx}_{I_1} + \underbrace{\int_{-\infty}^{+\infty} \frac{e^{-ix}}{2(x^2-4x+1-4i)} dx}_{I_2}$$



$$\gamma: \mathbb{C} \cup [-R; R]$$

$$I_1^*: \int_{\gamma} f(z) dz, \quad f(z) = e^{iz} \cdot \varphi(z)$$

$$\varphi(z) = \frac{1}{2} \frac{z^2}{z^2-4z+1-4i}$$

$$\mathcal{C}_R = \{z \mid |z|=R, \text{Im } z > 0\}$$

$$\text{No residue at } z=0: \int_{\mathcal{C}_R} e^{iz} \varphi(z) dz \rightarrow 0, R \rightarrow +\infty$$

$$\text{r.f. } \alpha = 1 > 0, \text{ so } |\varphi(z)| \sim \frac{1}{R^2} \rightarrow 0, R \rightarrow \infty$$

$$\int_{-R}^R f(x) dx \rightarrow I_1, R \rightarrow \infty \Rightarrow I_1^* \rightarrow I_1, R \rightarrow \infty$$

$$(z^2-4z+1-4i): \quad \mathcal{Q} = a+bi, \quad \sqrt{\mathcal{Q}} = (a+bi)^2 \Rightarrow$$

$$\Rightarrow a=4, b=2 \Rightarrow \begin{cases} z_1 = 4+i \\ z_2 = -i \end{cases}$$

$$I_1^* = 2\pi i \text{Res}_{z=4+i} f(z) = 2\pi i \cdot \frac{e^{-z}}{2(z-4)} \Big|_{z=4+i} = \frac{\pi e^{4i-1}}{4+2i} = \frac{\pi(2-i)e^{4i-1}}{10}$$

$$= \frac{\pi i (\cos 4 + i \sin 4)(2-i)}{e \cdot 10} = \frac{\pi i (2 \cos 4 + \sin 4) + \pi (\cos 4 - 2 \sin 4)}{10e}$$

$$I_2 = 2\pi i \text{Res}_{z=-i} f(z), \quad z = -2 \pm (2+i) \xrightarrow{-4-i} i$$

$$\text{ar: } \frac{e^{iz}}{2(z+4)} \Big|_{z=-i} = \frac{\pi \cdot e^{-1}}{2i+4} = \frac{\pi(4-2i)}{20e} = \frac{\pi+2\pi i}{10e}$$

$$f = \frac{\pi}{10e} (\cos 4 + 1 - 2 \sin 4) +$$

$$+ \frac{\pi i}{10e} (2 \cos 4 + 2 + \sin 4)$$

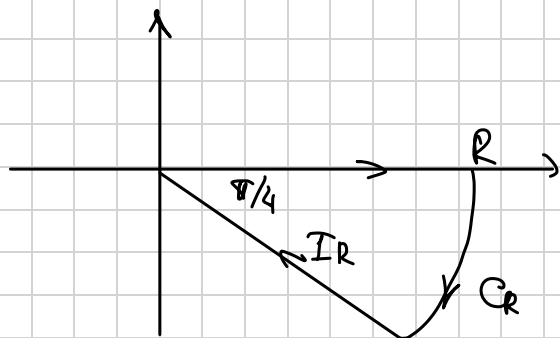
12 $I = \int_0^{\infty} e^{-x^2} dx \quad \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

Change: $t^2 = -x^2 \Rightarrow t = x e^{\frac{\pi i}{4} + \pi k i} \Rightarrow x = t e^{-\frac{\pi i}{4}}$

$\gamma_R = [0, R] \cup C_R \cup I_R$

$C_R = \{z \mid z = R e^{i\varphi}, \varphi \in [0, -\pi/4]\}$

$I_R = \{z \mid z = t e^{-i\pi/4}, t \in [R, 0]\}$



$f(z) = e^{-z^2} \quad \int_{\gamma_R} f(z) dz = 0$

$z \in C_R : z^2 = R^2 e^{2i\varphi} = R^2 (\cos 2\varphi + i \sin 2\varphi), \varphi \in [0, -\pi/4] \Rightarrow$

$\Rightarrow |e^{-z^2}| = e^{R^2 \sin 2\varphi} \Rightarrow \left| \int_{C_R} e^{-z^2} dz \right| \leq \int_0^{\pi/4} R e^{-R^2 \sin 2\varphi} d\varphi \stackrel{\varphi \rightarrow \frac{\varphi}{2}}{=} \frac{1}{2} \int_0^{\pi/2} R e^{-R^2 \sin \varphi} d\varphi \leq \frac{R}{2} \int_0^{\pi/2} e^{-R^2 \frac{2}{\pi} \varphi} d\varphi = \frac{R}{2} \frac{\pi}{2R^2} (1 - e^{-R^2}) \leq \frac{\pi}{4R} \xrightarrow{R \rightarrow \infty} 0$

$\int_{I_R} f(z) dz = \int_R^0 e^{-t^2} e^{-i\pi/4} dt \xrightarrow{R \rightarrow \infty} - \int_0^{\infty} e^{-t^2} dt e^{-i\pi/4} = -\frac{\sqrt{\pi}}{2} e^{-i\pi/4}$

$\begin{cases} z = t e^{-i\pi/4} \\ i z^2 = t^2 \end{cases}$

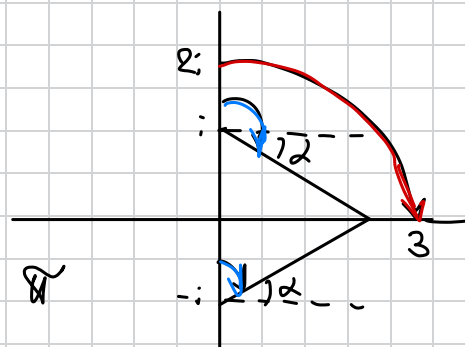
$0 = \int_{\gamma_R} f(z) dz = \int_{C_R} + \int_{I_R} + \int_0^R \Rightarrow I = \frac{\sqrt{\pi}}{2} e^{-i\pi/4}$

6.3. $f(z) = z^2 + 1 \quad \gamma: z(t) = x(t) + i y(t)$

$z(0) = 2i \quad z(\frac{\pi}{2}) = 3$

$x(t) = 2 \sin t \quad y(t) = 2 \cos t \quad t \in [0, \pi/2]$

$\Delta \arg f(z) = \Delta \arg(z-i) + \Delta \arg(z+i) = \frac{\pi}{2} + \alpha + \frac{\pi}{2} - \alpha = \pi$

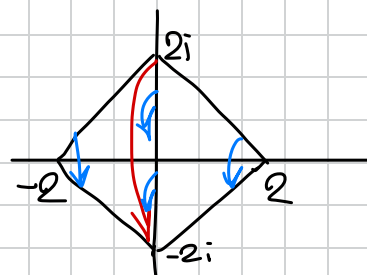


6.5. $f(z) = \frac{z^2 + 1}{z^2 - 4}$

$\gamma: z(t) = x(t) + i y(t)$

$x(t) = \cos t$

$y(t) = 2 \sin t$



$$z(\pi/2) = 2i \quad z(3\pi/2) = -2i \quad \Delta \arg f(z) = \pi + \pi + \alpha - \alpha = 2\pi$$

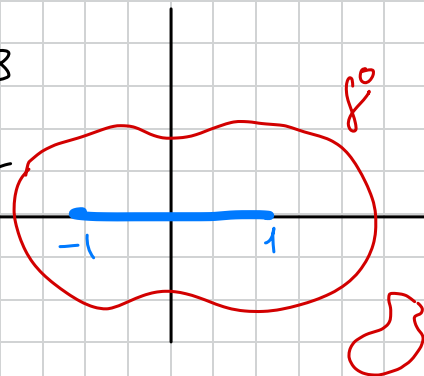
17.4. $\sqrt{z^2 - z}$ $\mathcal{D} = \mathbb{C} \setminus [-1, 1]$

Пусть $z \in \mathcal{D}$ - произвольная к-г. точка

1. $[-1, 1]$ выше z : $\Delta \arg f(z) = 2\pi + 2\pi = 4\pi : 2\pi$

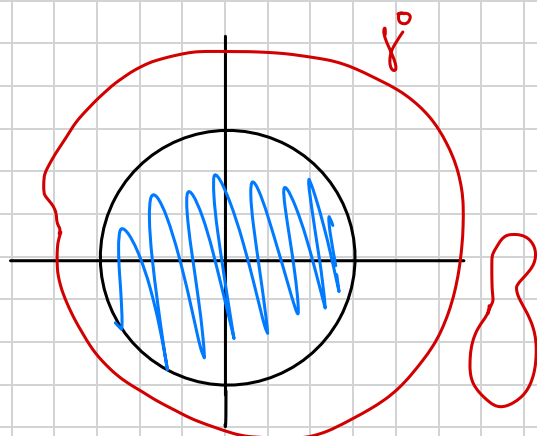
2. $[-1, 1]$ ниже z : $\Delta \arg f(z) = 0 : 2\pi$

Вот р. б. учас.



17.6. $\ln \frac{z^2 - 1}{z^2 + 1}$ $\mathcal{D} = \{z \mid |z| > 2\}$

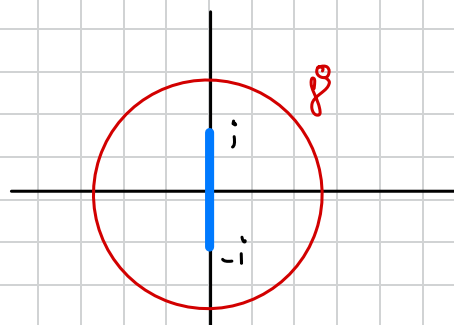
высше $2\pi + 2\pi - 2\pi - 2\pi = 0$
ниже $0 + 0 - 0 - 0 = 0 \Rightarrow \exists$



17.7. $\sqrt[3]{1+z^2}$ $\mathcal{D} = \mathbb{C} \setminus [-i; i]$

$\mathcal{D} = \{z \mid |z| = 2\} \subset \mathcal{D}$

$\Delta \arg = 2\pi + 2\pi = 4\pi \neq 6\pi \Rightarrow \nexists$



18.9. $\sqrt[3]{1-z^2}$ $\varphi(0)=1$, $\varphi(-3)=?$

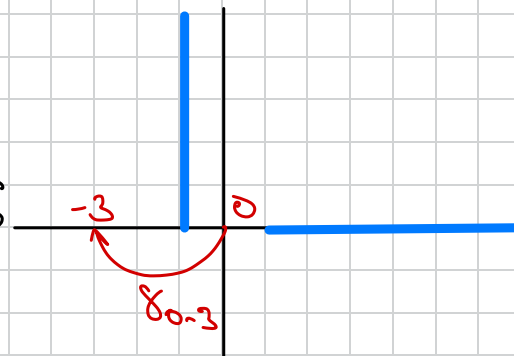
1) $\mathcal{D} = \mathbb{C} \setminus ([1, +\infty] \cup [-1, -1+i\infty])$

Так $\mathcal{D}$ произв. открыт. к-г. точка $z \in \mathcal{D}$

$\Delta \arg(1-z^2) = \Delta \arg(1-z) + \Delta \arg(1+z) = 0 \Rightarrow \text{р.б.} \exists$

$\varphi(-3) = \varphi(0) \sqrt{\frac{f(-3)}{f(0)}} e^{i/3 \Delta \arg f(z)} = \sqrt[3]{8} e^{-i\pi/3} = 1 - i\sqrt{3}$

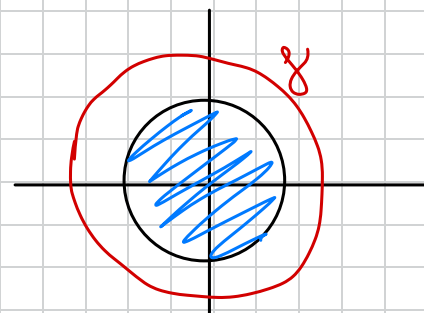
2) Аналогично $\varphi(-3) = 1 - i\sqrt{3}$



18.20. 2) $\{2\ln(z+1) - \ln(z-i)\}$ $G = \{z: 1 < |z| < \infty\}$

$$\Delta_8 \arg f(z) = 2\Delta_8 \arg(z+1) - \Delta_8 \arg(z-i) =$$

$$= 2 \cdot 2\pi - 2\pi = 2\pi \neq 0 \Rightarrow \nexists \text{ p.b.}$$



18.24. $\ln z$. $\psi(1) = 0$

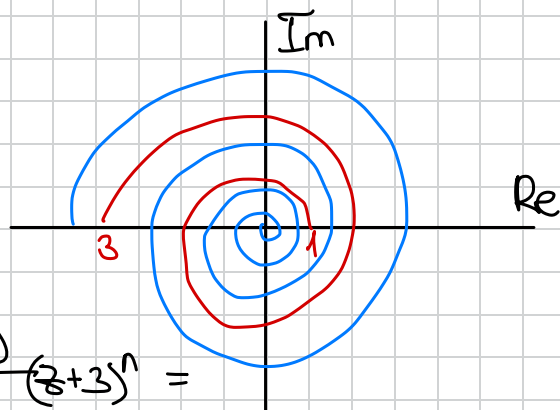
1) $\psi(2) = \frac{f'(2)}{f(2)} = \frac{1}{2}$

2) $\psi'(z) = \frac{1}{z}$ $\psi''(z) = -\frac{1}{z^2}$, ..., $\psi^{(n)}(z) =$

$$= \frac{(-1)^{n-1} (n-1)!}{z^n} \Rightarrow \psi(z) = \psi(-3) + \sum_{n=1}^{\infty} \frac{\psi^{(n)}(-3)}{n!} (z+3)^n =$$

$$= \psi(-3) + \sum_{n=1}^{\infty} \frac{-1}{3^n n} (z+3)^n, \quad \psi(-3) = \psi(1) + \ln\left|\frac{-3}{1}\right| + i\Delta_8 \arg z = \ln 3 + 3i\pi$$

$$\Rightarrow \psi(z) = \ln 3 + 3i\pi - \sum_{n=1}^{\infty} \frac{(z+3)^n}{n \cdot 3^n}$$



18.25. $\sqrt[3]{z}$ $1 \leq t \leq 3$ $t = z-1$

$\varphi(-1) = -1$

1) $\varphi(-2) = \varphi(-1) \sqrt[3]{\left|\frac{-2}{-1}\right|} e^{-i/3 \Delta_8 \arg z} = \sqrt[3]{2} e^{\frac{2i\pi}{3}}$

2) $\varphi'(z) = \frac{f'(z)}{3\varphi(z)^2}$ $\varphi(-3) = \varphi(-1) \sqrt[3]{3} e^{i/3 \cdot 2\pi} = \sqrt[3]{3} e^{\frac{4i\pi}{3}} \Rightarrow$

$$\Rightarrow \varphi'(-3) = 3^{-5/3} e^{-\frac{8i\pi}{3}} = 3^{-5/3} e^{-\frac{2i\pi}{3}}$$

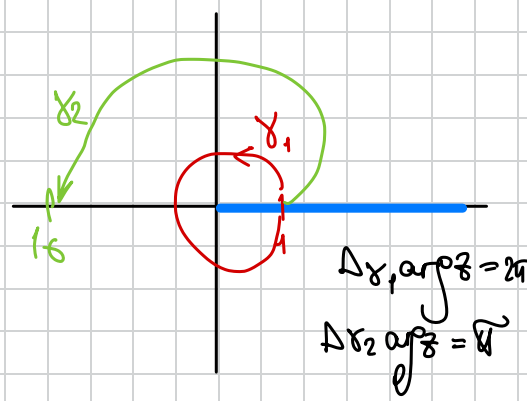
3) $\varphi(z) = \varphi(2) \sum_{n=0}^{\infty} C_{1/3}^n \frac{(z-2)^n}{2^n} = \varphi(2) = \varphi(-1) \sqrt[3]{2} e^{i/3 \cdot 3\pi} = \sqrt[3]{2} \sum_{n=0}^{\infty} C_{1/3}^n \frac{(z-2)^n}{2^n}$

18.27. $\varphi(1+i0) = 1$

1) $\varphi(1-i0) = \varphi(1+i0) \sqrt[4]{\left|\frac{1-i0}{1+i0}\right|} e^{i/4 \cdot 2\pi} = i$

2) $\varphi(16-i0) = \varphi(1+i0) \sqrt[4]{\left|\frac{16-i0}{1+i0}\right|} e^{i/4 \cdot 2\pi} = 2i$

3) $\varphi(-16) = \varphi(1+i0) \sqrt[4]{\left|\frac{-16}{1+i0}\right|} e^{i/4 \cdot \pi} = 2e^{i\pi/4}$



$$4) g'(z) = \frac{1}{4g(z)^3} \Rightarrow g'(-6) = \frac{1}{32e^{3i\pi/4}}$$

$$5) g''(z) = \frac{-3}{4g(z)^4} g'(z) \Rightarrow g''(-6) = \frac{3}{2^4 e^{3i\pi/4}}$$

18.35. $g(z) = \sqrt[3]{z+9}^* = (\sqrt[3]{z+9} \cdot e^{\frac{i}{3} \Delta \arg(z+9)})e$

Paras $z = 9e^{it} \quad -\pi \leq t \leq \pi/2$

Wz $z = 9 + ti, \quad t \geq 0$

$$\begin{cases} z = -9 \\ z = \infty \end{cases} \quad e^{\frac{i}{3}(\cdot 0)} \cdot C \sqrt[3]{z+9} = C \sqrt[3]{z+9} \Rightarrow$$

$\Rightarrow$ maximal Kong. per. Berks

Größte kongruentes Berks

$\arg(g(10)) = \frac{2\pi}{3}$

$$g(10) = \sqrt[3]{10+9} \cdot e^{\frac{i}{3}(\cdot 0)} \cdot C$$

$$\varphi = \frac{2\pi}{3} \quad |C| \cdot \sqrt[3]{z+9} = |g| |C| e^{i\varphi}$$

Per. Berks: $\sqrt[3]{z+9} \cdot e^{i/3 (\Delta \arg(z+9) + 2\pi)}$

1) $g(-8), g(-1), g'(0), g(-9+8i)$

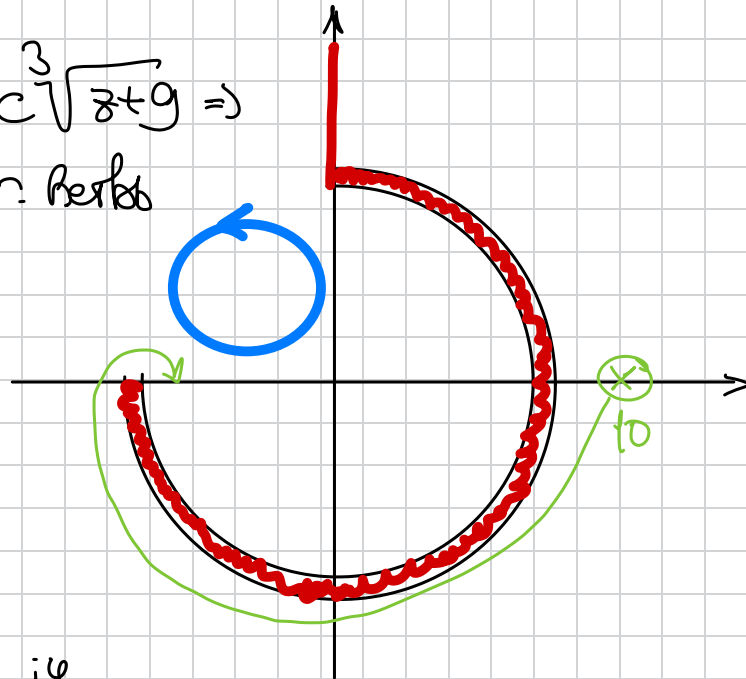
$$g(-8) = 1 \cdot e^{3(-2\pi)} \cdot 1 \cdot e^{i \frac{2\pi}{3}} = 1$$

$$g(-1) = 2e^{i/3(-2\pi)} \cdot 1 \cdot e^{i \frac{2\pi}{3}} = 2$$

$$g(-9+8i) = 2e^{i/3(2\pi - 3/2\pi)} = 2e^{i\pi/6}$$

$g^3 = z+9 \quad \frac{d(z+9)}{dz} = 3g^2 g' = 1 \Rightarrow g' = \frac{1}{3g^2}$

$$g(0) = \sqrt[3]{9} e^{i/3(2\pi - 2\pi)} \Rightarrow g'(0) = \frac{1}{3 \cdot 9^{2/3}}$$



$$g(z) = \sqrt[3]{12} e^{i/3(2\pi - 2\pi)} = \sqrt[3]{12}$$

$$t = z - 3, z = t + 3$$

$$g(t+3) = \sqrt[3]{t+12} \cdot e^{i/3(\arg(t+12) + 2\pi)}$$

$$u = x - 3 \Rightarrow g(u+3) = \sqrt[3]{u+12} e^0 = \sqrt[3]{12} \sqrt[3]{1 + \frac{u}{12}} =$$

$$= \sqrt[3]{12} \sum_{k=0}^{\infty} \left(\frac{u}{12}\right)^k C_{1/3}^k$$

$$g(z) = \sum_{k=0}^{\infty} C_{1/3}^k \left(\frac{z-3}{12}\right)^k \sqrt[3]{12} \quad \left|\frac{z-3}{12}\right| < 1, |z-3| < 12$$

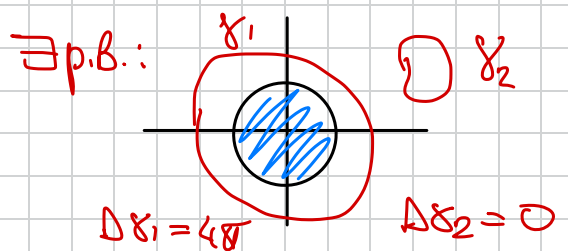
9.1.4) $I = \oint_{\gamma} \frac{dz}{g_k(z)}$, $g_k(z) \in \sqrt{4z^2 + 4z + 3}$, $\mathcal{Q} = \{z : |z| > 1\}$

$$g_k(z) = C_k \sqrt{|4z^2 + 4z + 3|} e^{i/2(\Delta \arg f(z))}, k \in \{0, 1\}$$

$$g_k(0) = \sqrt{3} e^{i/2 \Delta \arg k}, C_0 = 1, C_1 = -1 \quad z = \infty - \text{cross}$$

$$\text{res}_{z=\infty} \frac{i}{g_k(z)} = -C_{-1} = \begin{cases} -1/2, k=0 \\ 1/2, k=1 \end{cases} \quad z = -\frac{1}{2} \pm \frac{\sqrt{5}}{2} : \notin \mathcal{Q}$$

$$\oint_{\gamma} \frac{dz}{g_k(z)} = -\left(\text{res}_{z=\infty} \frac{1}{g_k(z)}\right) 2\pi i = (-1)^k \pi i$$

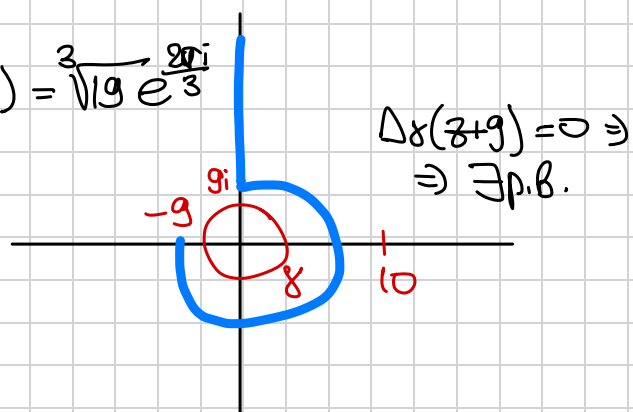


$$\frac{1}{g_k(x)} = \frac{1}{C_k \sqrt{4x^2 + 4x + 3}} = \frac{C_k}{2x} \frac{1}{\sqrt{1 + \frac{1}{x} + \frac{3}{4x^2}}} = \frac{C_k}{2x} + o\left(\frac{1}{|x|}\right)$$

19.5. $\sqrt[3]{z+9}$ $\arg f(10) = \frac{2\pi}{3} \Rightarrow f(10) = \sqrt[3]{9} e^{i\frac{2\pi}{3}}$

$$\oint_{|z|=5} \frac{2-f(z)}{(z+1)^2} dz = \oint$$

Wort: $z = -1$ - Breyer



$$f(-1) = 2 \text{ wg 18.35.}, \quad f'(-1) = \frac{1}{3f(-1)} = \frac{1}{12} \Rightarrow z = -1 - i\pi^1$$

$$\operatorname{res}_{z=-1} \frac{2-f(z)}{(z+1)^2} = \lim_{z \rightarrow -1} \frac{2-f(z)}{z+1} = -\frac{1}{12} \Rightarrow J = 2\pi i \operatorname{res}_{z=-1} \frac{2-f(z)}{(z+1)^2} = \frac{-\pi i}{6}$$

19.10. $\sqrt[3]{z-1}$ $f(1-) = ;$

$$\oint \frac{z(f(z)-2)}{(z-9)^2} dz = J$$

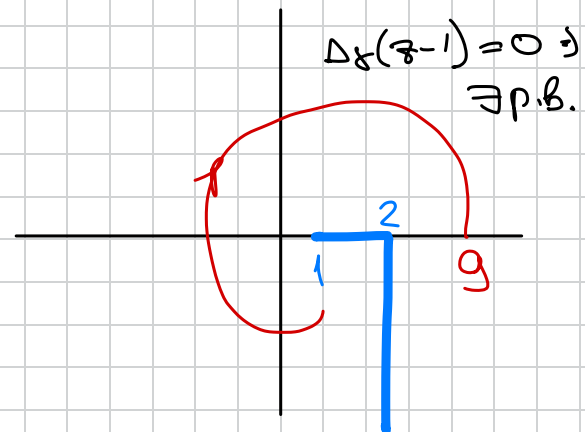
$|z-9|=2$

Wort x: $z=9$

$$f(9) = f(1-) \sqrt[3]{\left|\frac{8}{i}\right|} e^{i/3(-\frac{3\pi}{2})} = 2$$

$$f'(9) = \frac{1}{3f^2(9)} = \frac{1}{12} \Rightarrow z = 9 - i\pi^1 \Rightarrow$$

$$\Rightarrow \operatorname{res}_{z=9} \frac{z(f(z)-2)}{(z-9)^2} = \lim_{z \rightarrow 9} \frac{z(f(z)-2)}{z-9} = 9f'(9) = \frac{3}{4} \Rightarrow J = \frac{3\pi i}{2}$$



19.24. $\sqrt{2z^2+1}$ $f(0)=1$

$$\oint \frac{z dz}{\underbrace{(z+2)(f(z)+3)}_{g(z)}} = J$$

Wort x: $z=-2$ - free konvergenz

$$f(z) = -3 \Rightarrow z = \pm 2$$

$$f(2) = f(0) \sqrt{\left|\frac{2 \cdot 4 + 1}{1}\right|} e^{i\pi} = -3 - \text{neg x.}$$

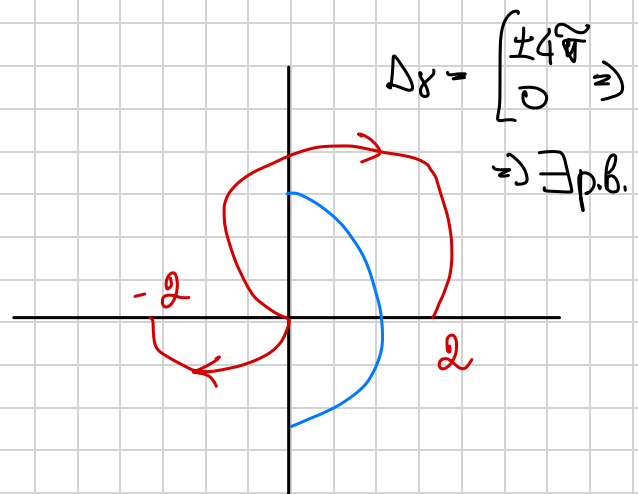
$$f(-2) = f(0) \sqrt{\left|\frac{2 \cdot 4 + 1}{1}\right|} e^{i0} = 3 - \text{pos.}$$

$$\operatorname{res}_{z=2} g(z) = \lim_{z \rightarrow 2} \frac{z}{(z+2)f'(2)} = \frac{1}{2f'(2)} = -\frac{3}{8}$$

$$\operatorname{res}_{z=-2} g(z) = -1/3$$

$$\forall x > 1/\sqrt{2}: f(x) = \sqrt{2x^2+1} \Rightarrow g(x) = \frac{x}{(x+2)(3-\sqrt{2x^2+1})}$$

$$\operatorname{res}_{\infty} g(z) = \lim_{\infty} z(g(\infty) - g(z)) = 1/\sqrt{2}$$



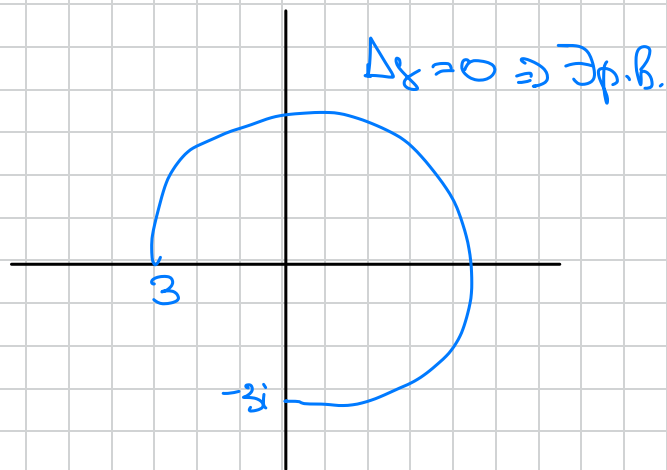
$$J = -2\pi i \left(\frac{1}{\sqrt{2}} - \frac{3}{8} - \frac{1}{3} \right) = \pi i \left(\frac{17}{12} - \sqrt{2} \right)$$

19.42. $\ln \frac{3+z}{iz-3} \quad h(\infty) = -\frac{5}{2}\pi i$

$$\oint_{|z|=1} \frac{dz}{(h^2(z)+\pi^2)^2}$$

Wortlich: $e^{h(z)} = e^{\pm \pi i} = -1 \Rightarrow$

$$\Rightarrow \frac{3+z}{iz-3} = -1 \Rightarrow z=0$$



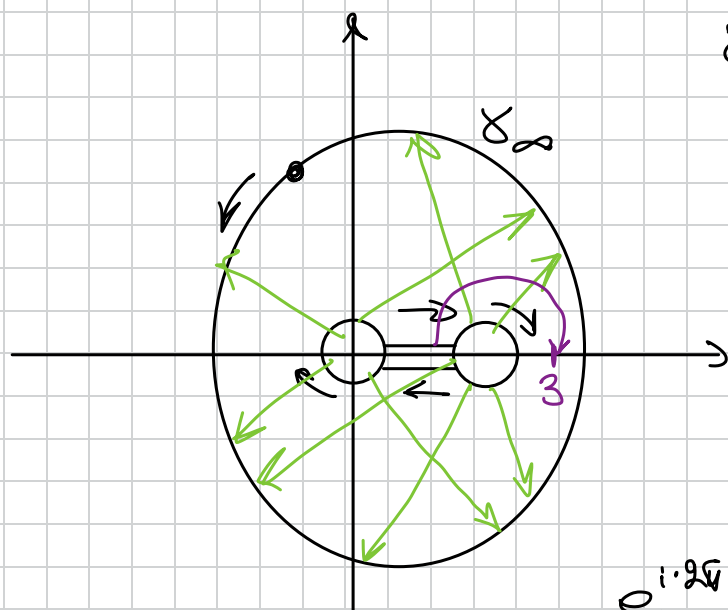
$$\forall x > 0 \quad h(x) = h(0) + \ln \left| \frac{3+x}{ix-3} \right| + i \Delta \gamma \arg \left(\frac{3+x}{iz-3} \right) =$$

$$= h(0) + \ln \frac{3+x}{\sqrt{x^2+9}} + i(-2\pi + \arctan \frac{x}{3}) \Rightarrow h(\infty) = h(0) + \ln 1 - \frac{3\pi i}{2} =$$

$$= -\frac{5}{2}\pi i \Rightarrow h(0) = -\pi i$$

23.5.3) $\int_0^1 \frac{dx}{(x-3)^2 \sqrt{x-x^2}}$

$$g(z) = \sqrt{z-z^2} * = C \sqrt{|z-z^2|} e^{i \overbrace{\Delta \gamma \arg x + \Delta \gamma \arg (x-1)}^{\Delta \gamma \arg (x(x-1))}}$$



$$\gamma = \gamma_1 \cup \gamma_2 \cup \gamma_3 \cup \gamma_4$$

$$\gamma_1 = [\varepsilon, 1-\varepsilon]$$

$$\gamma_2 = \{z \mid |z-1| = \varepsilon, \arg(z-1) \in [-\pi; \pi]\}$$

$$\gamma_3 = [1-\varepsilon, \varepsilon]$$

$$\gamma_4 = \{z \mid |z| = \varepsilon, \arg z \in [-\pi; \pi]\}$$

$$\Delta \gamma \arg(z(z-1)) = \dots = 2\pi + 2\pi$$

$$e^{i \cdot 2\pi} - 1 = 0 \Rightarrow \text{Beitrag hängen nicht zusammen}$$

Gen. hsp. Beibeh. $z_0 = \frac{1}{2} + i(0+0) = 0$

$$g(z_0) = \sqrt{x-x^2} = C \sqrt{|z-z^2|} C^{i/2 \Delta \gamma (\arg z + \arg(z-1))} \quad |C|=1$$

$$\downarrow \frac{1}{2} + i(0+0)$$

per Beibeh. $g(z) = \sqrt{|z-z^2|} e^{i/2 (\Delta \gamma \arg z + \Delta \gamma \arg(z-1))}$

$$I_\varepsilon = I_1 + I_2 + I_3 + I_4$$

$$I_1 = \int_{\varepsilon}^{1-\varepsilon} \frac{dx}{(x-3)^2 \underbrace{g(x+i0)}_{\sqrt{x-x^2}}} \rightarrow I, \varepsilon \rightarrow 0$$

$$I_3 = \int_{1-\varepsilon}^{\varepsilon} \frac{dx}{(x-3)^2 g(x-i0)} \rightarrow I, \varepsilon \rightarrow 0 \quad g(x-i0) e^{\frac{i}{2}(0-g\pi)} = -g(x+i0)$$

$$\oint_{|z|=\varepsilon} \frac{dz}{(z-3)^2 \sqrt{z-z^2}} e^{\frac{i}{2}(\Delta \arg z + \Delta \arg(z-1))}$$

$$z-1 = \varepsilon e^{i\varphi}$$

$$z = 1 + \varepsilon e^{i\varphi}$$

$$dz = \varepsilon i e^{i\varphi} d\varphi$$

αναλυτικό

$$I_4: \oint_{|z|=\varepsilon} \frac{dz}{(z-3)^2 \sqrt{z(z-1)}} e^{\frac{i}{2}(\Delta \arg z + \Delta \arg(z-1))}$$

$$z \sim \varepsilon e^{i0}$$

$$dz \sim \varepsilon A_1$$

$$z-3 \sim -3A_2$$

$$\sqrt{} \sim \sqrt{\varepsilon \cdot A_3}$$

$$I \sim \int \frac{\varepsilon}{\sqrt{\varepsilon}} A \rightarrow 0, \varepsilon \rightarrow 0$$

$$I_\varepsilon \rightarrow 2I, \varepsilon \rightarrow 0$$

$$C \text{ gr. crop. } I = 2\pi i \left(\operatorname{res}_{z=3} f(z) + \operatorname{res}_{z=\infty} f(z) \right)$$

$$f(z) = \frac{1}{(z-3)^2 g(z)}$$

$$\operatorname{res}_{z=3} f(z) = \operatorname{res}_{z=3} \frac{1}{(z-3)^2 g(z)}$$

$$g(z) = \sqrt{z} e^{i\frac{1}{2}(-\pi)} = \sqrt{z}(-i)$$

$$\operatorname{res}_{z=3} f(z) = \lim_{z \rightarrow 3} \frac{1}{g(z)} (f(z)(z-3)^2) = \lim_{z \rightarrow 3} \left(-\frac{1}{g(z)} g'(z) \right) \left[g^2 = z-z^2 = 6 \right]$$

$$2gg' = 1-2z \Rightarrow g' = \frac{1-2z}{2g} = \frac{-5}{2\sqrt{6}i}$$

$$\int \frac{1}{(z-3)^2 \sqrt{z(z-1)}} e^{i\frac{\pi}{2}} = \int \frac{1}{x^2(1-\frac{z}{x}) \sqrt{x(1-\frac{1}{x})(-i)}} \quad C_1 = 0 \Rightarrow \Rightarrow \operatorname{res}_{z=\infty} = 0$$

$$I_\varepsilon = 2I$$

$$I_\varepsilon = 2\pi i \left(\frac{-5i}{2\sqrt{6}} \right) \Rightarrow I = \frac{5\pi}{2\sqrt{6}}$$

$$4) \int_0^6 \frac{1}{x+2} \sqrt{\frac{x}{6-x}} dx = I$$

1. $\forall \varepsilon$ -np., 3., k.-r. $\Delta \arg z - \Delta \arg(6-z) = 0 \Rightarrow \exists \text{ p.b.}$

$$2. g(x+i0) = \sqrt{\frac{x}{6-x}} \quad \forall x \in [0, 6]$$

$$g(z) = \sqrt{\frac{z}{6-z}} e^{i/2 \Delta \arg z - \Delta \arg(6-z)}$$

Тогда на ~~резальном~~ ~~сегмент~~ ~~резальном~~ ~~сегмент~~ $g(x-i0) = \sqrt{\frac{x}{6-x}} e^{i/2 \cdot 2\pi} = -g(x+i0)$

$$3. \gamma = \gamma_+ \cup C_\varepsilon^0 \cup \gamma_- \cup C_\varepsilon^6$$

$$\text{Уточн: } z = -2, \quad g(-2) = i/2, \quad \text{res } f(z) = g(-2) = i/2$$

$$f(x) = \frac{1}{x+2} \sqrt{\frac{x}{6-x}} e^{i/2 \pi} = i \left(1 - \frac{2}{x} + \frac{4}{x^2} + \dots\right) \left(1 + \frac{3}{x} + \frac{3}{8} \frac{36}{x^2} + \dots\right) \Rightarrow$$

$$\Rightarrow \text{res } f(z) = -C_{-1} = -i$$

$$\left| \int_{C_\varepsilon^0} \frac{1}{z+2} g(z) dz \right| \leq \int |1| \leq \frac{1}{6} 2\pi \varepsilon \sqrt{\frac{\varepsilon}{6-\varepsilon}} \rightarrow 0$$

$$4. \oint_\gamma f(z) dz = \int + \int + \int + \int = 2\pi i \left(\frac{i}{2} - i\right) \Rightarrow I = \pi/2$$

$$7) \int_0^3 \frac{1}{(x+2)^2} \sqrt{\frac{1}{x^3(3-x)}} dx = I$$

1. $\forall \varepsilon$ -np., 3., k.-r. $3\Delta \arg z + \Delta \arg(3-z) = \begin{cases} 0 \\ 2\pi \end{cases} \Rightarrow \exists \text{ p.b.}$

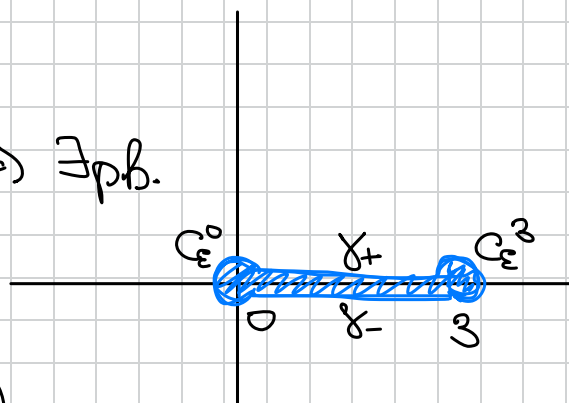
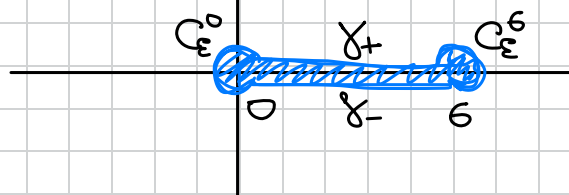
$$2. g(x+i0) = \sqrt[4]{\frac{1}{x^3(3-x)}} \quad \forall x \in [0, 3]$$

$$g(z) = \sqrt[4]{\frac{1}{z^3(3-z)}} e^{i/4 (3\Delta \arg z + \Delta \arg(3-z))}$$

Тогда на ~~резальном~~ ~~сегмент~~ ~~резальном~~ ~~сегмент~~ $g(x-i0) = g(x+i0) e^{\frac{3\pi i}{2}}$

$$3. \gamma = \gamma_+ \cup C_\varepsilon^0 \cup \gamma_- \cup C_\varepsilon^3$$

$$\text{Уточн: } z = -2, \quad g(-2) = \sqrt[4]{\frac{1}{8}} e^{\frac{3\pi i}{4}}$$



$$\operatorname{res}_{-2} f(z) = \left(\frac{1}{g(z)} \right)' (-2) = \frac{-17}{40^{5/4}} e^{\frac{\pi i}{4}}$$

$$f(x) = \frac{1}{x^3} \left(1 - \frac{1}{x} + \frac{12}{x^2} + \dots \right) \left(1 + \frac{1}{2x^4} + \dots \right) \Rightarrow \operatorname{res}_{\infty} f(z) = -c_1 = 0$$

$$\left| \int_{C_3} \right| \leq \int_{C_3} 1 \leq \frac{1}{C_3} \frac{2\pi\epsilon}{\sqrt[4]{\epsilon^3(3-\epsilon)}} \rightarrow 0, \epsilon \rightarrow 0, \text{ und } C^0 \text{ annahm.}$$

$$4. \oint f(z) dz = \int + \int + \int + \int = I + 0 - I e^{\frac{3\pi i}{2}} + 0 \Rightarrow I = \frac{17}{4 \cdot 10^{5/4}} \pi$$

23.6. 4) $\int_0^{+\infty} \frac{\ln x}{(x+1)^4 \sqrt{x^3}} dx = I$

1. Das $\{ \ln z \}$ u. $\{ \sqrt[4]{x^3} \}$

haben Fortsetzungen along. reifgezogen $\Rightarrow$

$\Rightarrow \exists$ per. Fortsetz.

2. $g(x+i0) = \sqrt[4]{x^3} \quad \forall x \in [0, R]$

$$g(z) = \sqrt[4]{z^3} e^{i \frac{1}{4} \arg z}$$

$$h(x+i0) = \ln x \quad \forall x \in [0, R]$$

$$h(z) = \ln z + i \frac{1}{4} \arg z$$

Nach der ~~unendlichen~~ d. reifgez. $g(x-i0) = \sqrt[4]{x^3} e^{\frac{3\pi i}{2}}$

$$h(x-i0) = \ln x + 2\pi i$$

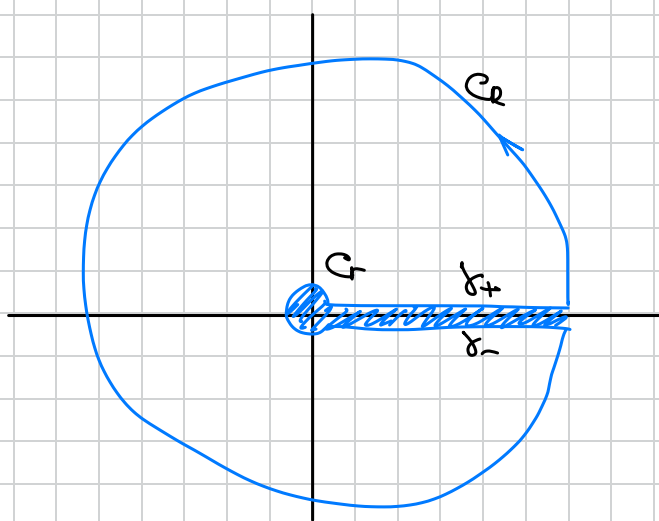
3. Wert: $z = -1$, $\operatorname{res}_{-1} f(z) = \frac{h(-1)}{g(-1)} = \pi i e^{-\frac{3\pi i}{4}}$

$$\left| \int_{C_R} f(z) dz \right| \leq \frac{\ln R + \pi}{(R-1)^4 \sqrt[4]{R^3}} \cdot 2\pi R \rightarrow 0, R \rightarrow \infty$$

$$\left| \int_{C_r} f(z) dz \right| \leq \frac{\ln r + \pi}{(1-r)^4 \sqrt[4]{r^3}} \cdot 2\pi r \rightarrow 0, r \rightarrow 0$$

$$4. \oint = \int + \int + \int + \int = 2\pi i \cdot \pi i e^{-\frac{3\pi i}{4}} \Rightarrow \begin{cases} I + 0 - I e^{\frac{-3\pi i}{2}} - I' + 0 = -2\pi^2 e^{\frac{-3\pi i}{2}} \\ I' + 0 - I' e^{\frac{-3\pi i}{2}} + 0 = -2\pi^2 e^{\frac{-3\pi i}{2}} \end{cases} \Rightarrow$$

$$\Rightarrow I = -\sqrt{2} \pi^2$$



23.6. 6) $\int_1^{+\infty} \frac{1}{(x^2+1)\sqrt{x-1}} dx = I$

1. Das $\sqrt{x-1}$ 1 u ∞

haben Fortsetzungs vollg. gezogen $\Rightarrow$

$\Rightarrow \exists$ per. Fortset.

2. $g(x+i0) = \sqrt{x-1} \quad \forall x \in [1, \infty]$

$g(z) = \sqrt{z-1} e^{\frac{i}{2} \Delta \arg(z-1)}$

Nach der ~~unendlichen~~ d. ~~Fortsetzung~~ $g(x-i0) = \sqrt{x-1} e^{i/2 \cdot 2\pi} = i g(x+i0)$

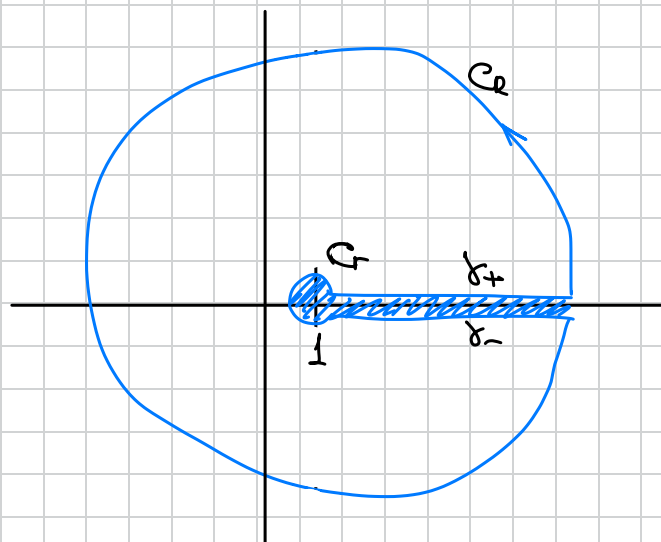
3. Werten: $z = \pm i$, $\text{res } f(z) = \frac{1}{z \cdot g(z)} = \frac{1}{2i} (\sqrt{2} e^{i/2 \cdot 3\pi/4})^{-1} = \frac{-i}{2\sqrt{2}} e^{-3\pi i/8}$

$\text{res } f(z) = \frac{1}{-z \cdot g(-z)} = \frac{1}{2\sqrt{2}} e^{-5\pi i/8}$

$\left| \int_{C_R} f(z) dz \right| \rightarrow 0, R \rightarrow \infty$

$\left| \int_{C_r} f(z) dz \right| \rightarrow 0, r \rightarrow 0$

4. $\oint = \int + \int + \int + \int = 2\pi i \cdot \frac{-i}{2\sqrt{2}} (e^{-3\pi i/8} - e^{-5\pi i/8}) \Rightarrow I - iI = \frac{\pi}{\sqrt{2}} (e^{-3\pi i/8} - e^{-5\pi i/8})$



23.6. 8) $\int_0^{+\infty} \frac{x \ln x}{(x+1)(x+2)} dx = I$

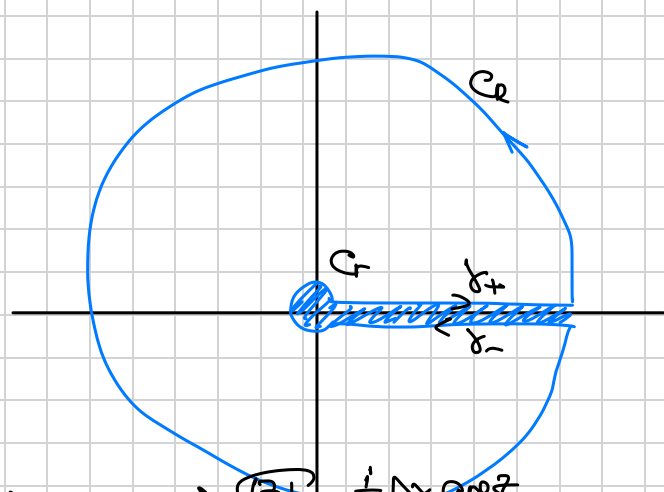
1. $\Delta \arg z = 0 \quad \forall z \in \mathbb{C} \Rightarrow$

b $\mathbb{C} \exists$ per. b. $\sqrt{z}$ u $\ln z$

2. $g(x+i0) = \sqrt{x}$

$g(x-i0) = \sqrt{x} e^{i/2 \cdot 2\pi} = -\sqrt{x}$

$g(z) = g(1+i0) \sqrt{\frac{z-1}{1}} e^{\frac{i}{2} \Delta \arg z}$



$$h(x+i0) = \ln x$$

$$h(x-i0) = \ln x + 2\pi i$$

$$g(-1) = i \quad g(-2) = i\sqrt{2} \quad h(-1) = i\pi \quad h(-2) = \ln 2 + i\pi$$

$$3. \text{ Res } f(z) = \frac{g(z)h(z)}{(z+2)} \Big|_{z=-1} = -\pi$$

$$z = -2 - \pi^2 \Rightarrow \text{Res } f(z) = \ln(2) + i\pi$$

$$\left| \int_{\Gamma} \right| \leq \int_0^{2\pi} \frac{r |\ln r + i\varphi|}{(1-r)(2-r)} r d\varphi \leq 4r^{3/2} \rightarrow 0, r \rightarrow 0$$

$$\left| \int_{\Gamma_R} \right| \leq \frac{B}{\sqrt{R}} \rightarrow 0, R \rightarrow \infty$$

$$4. \text{ Res } f(z) = 2\sqrt{2} \pi \ln 2 + 2\pi^2 i(\sqrt{2}-1) = \int_+ + \int_- + \int_{\Gamma} + \int_{\Gamma_R} = \oint_{\gamma}$$

$$\int_+ \rightarrow I, \int_- \rightarrow I + 2\pi i I^*, I^* = \int_0^{+\infty} \frac{x dx}{(x+1)(x+2)} = (\sqrt{2}-1)\pi \Rightarrow$$

$$\Rightarrow I = 2\pi \ln 2$$

23.7.3)

$$I = \int_0^{+\infty} \frac{\ln x dx}{(x^2+1)^2}$$

$$\text{Recall. } J = \int_0^{+\infty} \frac{\ln^2 x dx}{(x^2+1)^2}$$

$$\Delta_{\gamma} \arg z = 0 \quad \forall \gamma \subset \mathcal{D} \Rightarrow \exists p.f.$$

$$h(x+i0) = \ln x$$

$$h(x-i0) = \ln x + 2\pi i$$

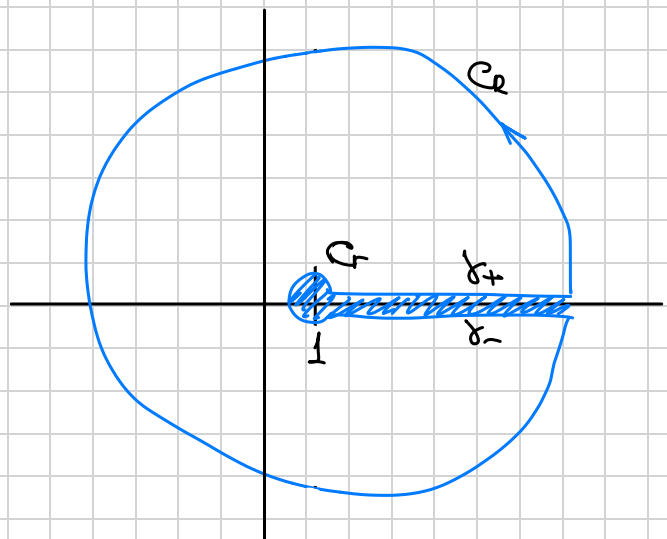
$$h(z) = \ln 1 + \ln \left| \frac{z}{1} \right| + i \Delta \arg z$$

$$h(i) = \frac{i\pi}{2}$$

$$h(-i) = \frac{3i\pi}{2}$$

$$\text{Res } f(z) = \lim_{z \rightarrow i} \left(\frac{f(z)}{(z+i)^2} \right)' = -\frac{\pi}{4} + \frac{i\pi^2}{16}$$

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$$z = -i - \pi^2, \quad \operatorname{Res}_z f(z) = \frac{3\pi}{4} - \frac{i \cdot 9\pi^2}{16}$$

$$2\pi i \operatorname{Res} = \pi^3 + \pi^2 i = \int + \int + \int + \int = \oint_\gamma$$

$$\left| \int_\gamma \right| \leq \int_0^{2\pi} \frac{|\ln r + i\varphi|^2}{(1-r^2)^2} r d\varphi \leq Ar \rightarrow 0, \quad r \rightarrow 0$$

$$\left| \int_\gamma \right| \leq \frac{B}{R^3} \rightarrow 0, \quad R \rightarrow \infty$$

$$\int_+ \rightarrow \mathcal{J}, \quad \int_- = \int_R^r \frac{(\ln x + 2\pi i)^2}{(x^2 + 1)^2} dx \rightarrow -\mathcal{J} - 4\pi i \mathcal{I} + \pi^3 \Rightarrow$$

$$\Rightarrow \pi^3 - 4\pi i \mathcal{I} = \pi^3 + \pi^2 i \Rightarrow \mathcal{I} = \frac{\pi^2 i}{-4\pi i} = -\frac{\pi}{4}$$

15.1. 3) $z^7 - 5z^4 + z^2 - 2 = 0$ $\mathcal{D} = \{z \mid |z| < 1\}$

$$|z^7|_\gamma = 1$$

$$\gamma = \{z \mid |z| = 1\}$$

$$|-5z^4|_\gamma = 5$$

$$|z^2|_\gamma = 1$$

$$|-2|_\gamma = 2$$

Пытаю $f(z) = -5z^4$, $g(z) = z^7 + z^2 - 2$, тогда не фн. Попробуем

$h(z) = f(z) + g(z)$ им. 4 корня в $\mathcal{D}$ как у $f(z)$

6) $z^4 - 9z + 1 = 0$ $\mathcal{D} = \{z \mid |z| < 2\}$

$$|z^4|_\gamma = 16$$

$$\gamma = \{z \mid |z| = 2\}$$

$$|9z|_\gamma = 18$$

$$|1|_\gamma = 1$$

Пытаю $f(z) = -9z$, $g(z) = z^4 + 1$, тогда не фн. Попробуем $f+g$ им 1 корень в $\mathcal{D}$ как у f

7) $z^6 - 6z + 10 = 0$ $\mathcal{D} = \{z \mid |z| > 1\}$

$$|z^6|_\gamma = 1$$

$$\gamma = \{z \mid |z| = 1\}$$

$$|-6z|_\gamma = 6$$

$$11018 = 10$$

Нужно $f(z) = 10$, $g(z) = z^6 - 5z$, но фн. Рунге у $f+g$ нет
коэффициентов в $\mathbb{C} \setminus \mathbb{R} \Rightarrow$ в $\mathbb{R}$ и $\mathbb{C}$ коэффициенты

$$10) \quad z^6 + 3z^4 + 2z^3 + 1 = 0 \quad \mathbb{B} = \{z \mid |z| < 2\}$$

$$|z^6|_8 = 64 \quad \delta = \{z \mid |z| = 2\}$$

$$|3z^4|_8 = 48$$

$$|2z^3|_8 = 16$$

$$|1|_8 = 1$$

Нужно $f(z) = z^6 + 1$, $g(z) = 3z^4 + 2z^3$. Но фн. Рунге у $f+g$
в коэффициентах в $\mathbb{R}$, как и у f .

15.4. Доказать, что при $k > 1$ упр-е $z = k - e^{-z}$ имеет в
 $\{z \mid \operatorname{Re} z \geq 0\}$ ровно 1 решение. Копируем

$$\square \text{ Нужно } f(z) = z - k, \quad g(z) = e^{-z}$$

Задача. $R > k + 1$, нужно $\gamma_R = [-iR, iR] \cup C_R$

$$\text{На } \text{ex. } z = iy \Rightarrow |f(z)| = |iy - k| \geq k > 1$$

$$|g(z)| = e^{-iy} = 1$$

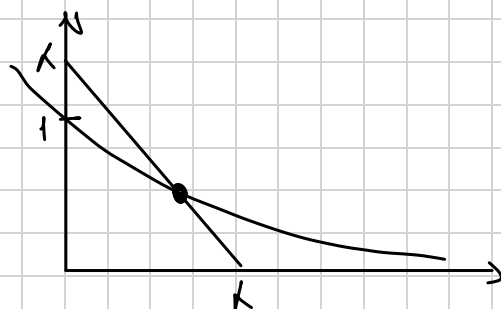
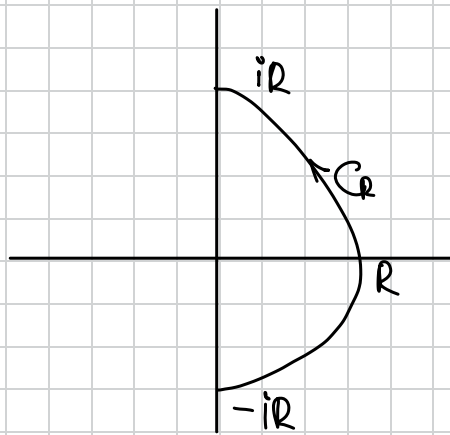
$$\text{На } C_R: z = Re^{i\varphi}, \quad \varphi \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow$$

$$\Rightarrow |f(z)| = |z - k| \geq |z| - k = R - k > 1$$

$$|g(z)| = e^{-x} \leq 1 \quad \forall x. \quad x \geq 0$$

Значит $\forall z \in \mathbb{R} \quad |f(z)| > |g(z)| \Rightarrow$ по фн. Рунге $f+g$ не
отомыкает коэффициенты, это и f ($z = k$)

$$\exists x_0 > 0: k - x_0 = e^{-x_0} \Rightarrow x_0 \in \mathbb{R} \quad \square$$



14. $2z^6 + 2z^3 - 5z - 2$ $G = \{z \mid |z| < 1\}$

$|2z^6|_G = 2$

$\gamma = \{z \mid |z| = 1\}$

$|2z^3|_G = 2$

$|-5z|_G = 5$

$|-2|_G = 2$

пусть $f(z) = -5z - 2$, $g(z) = 2z^6 + 2z^3$

по th. Руше g и $f+g$ 1 корень в G как у f

15. $\oint_{|z|=1} \sin\left(\frac{iz^3}{4z^3 - 2iz^2 + i}\right) dz$

Находим корни, $z = \infty$, z_1, z_2, z_3 — все значения

$4z^3 - 2iz^2 + 1 = 0$

$|4z^3|_G = 4 \Rightarrow f(z) = 4z^3$

$|2iz^2|_G = 2$
 $|1|_G = 1 \Rightarrow g(z) = 2iz^2 + 1$

$\oint \sin\left(\frac{iz^3}{4z^3 - 2iz^2 + 1}\right) dz = -2\pi i \operatorname{Res}_{z=\infty} f(z) = -2\pi i (-C-1) = 2\pi i \left(-\frac{1}{8} \cos \frac{1}{4}\right)$

$\sin \frac{i}{4} \left(1 + \frac{1}{2z} + o\left(\frac{1}{z}\right)\right) = \sin\left(\frac{i}{4}\right) \cos\left(-\frac{1}{2z} + o\left(\frac{1}{z}\right)\right) + \cos\left(\frac{i}{4}\right) \sin\left(-\frac{1}{2z} + o\left(\frac{1}{z}\right)\right)$
 $\Rightarrow C-1 = -\frac{1}{8} \cos \frac{1}{4}$